

CHAPTER ONE: INTRODUCTION

[REVISED VERSION — Addressing supervisor comments: misalignment, epistemological stance, problem statement, and research gap]

School plays a critical role in shaping a child's cognitive, social, emotional, and academic development. In South Africa, children spend a substantial portion of their waking hours in educational settings, making these environments central to their overall development. Learning institutions are designed not only to deliver structured education but also to support holistic growth through activities that foster creativity, physical development, and social skills (Eurokids Preschool, 2023).

Within this context, Early Childhood Development (ECD) centres are particularly significant, as they cater to children between birth and six years of age, a period that is foundational for long-term developmental outcomes. The physical environment of these centres, including indoor environmental quality and especially thermal conditions, plays a direct role in shaping children's health, comfort, behaviour, and learning experiences (Evans, 2006).

Climate change is increasingly intensifying environmental risks globally, including heatwaves and rising temperatures. In South Africa, warming trends have been recorded, with peri-urban areas such as Hammanskraal experiencing heightened exposure due to limited infrastructure and inadequate climate-control mechanisms. These conditions place young children at greater risk, as their thermoregulatory systems are not fully developed, making them more vulnerable to heat stress (UNICEF, 2023).

Research shows that poor indoor environmental conditions negatively affect attention, cognitive performance, and overall classroom behaviour, while also contributing to teacher fatigue and reduced instructional effectiveness (Mendell & Heath, 2005; Wargocki & Wyon, 2017). Local studies further indicate that classroom environments often fail to regulate heat effectively, with indoor temperatures closely reflecting outdoor conditions due to poor design and ventilation (Gibberd & Motsatsi, 2013).

Despite this, existing research has largely focused on formal school environments and older learners. There is limited attention given to ECD settings, particularly within peri-urban South African contexts, where infrastructure constraints and climate vulnerability intersect. Furthermore, the lived experiences and perceptions of ECD teachers, who directly observe and mediate children's responses to these conditions, remain underexplored. Their insights provide a critical lens for understanding how thermal stress influences behaviour, participation, and learning in real-world settings.

This study addresses this gap by focusing on classroom environment in Hammanskraal. It examines how thermal conditions affect the learning environment and how teachers perceive and respond to these challenges. Grounded in an interpretivist framework, the study prioritises teachers' lived experiences and subjective interpretations as valid forms of knowledge (Creswell & Poth, 2018; Bryman, 2016). By doing so, it establishes a foundation for the methodological approach that follows, which centres on qualitative data to explore the relationship between environmental conditions and pedagogical experiences.

1.2 Epistemological Stance

This study is grounded in an interpretivist epistemological framework. Rather than seeking to measure or model objective thermal conditions, the research investigates how ECD teachers perceive, interpret, and respond to elevated indoor temperatures in their centres. The central epistemological premise is that the meanings teachers assign to thermal discomfort — and the pedagogical and behavioural responses they adopt in response — constitute legitimate and valuable knowledge that cannot be captured by physical measurements alone (Creswell & Poth, 2018; Bryman, 2016).

The study therefore examines the interface between physical thermal exposure and its lived pedagogical consequences as perceived by teachers. This positions the research at the intersection of environmental health and phenomenological inquiry, making teachers' subjectivities — rather than instrumented temperature data — the primary unit of analysis.

1.3 Problem Statement

ECD centres in peri-urban South Africa, including those in Hammanskraal, frequently operate in buildings characterised by corrugated iron roofing, asbestos sheeting, and limited ventilation — materials and designs that amplify indoor temperatures during summer months. Yet policy frameworks governing ECD infrastructure in South Africa do not prescribe thermal comfort standards for these facilities, and no systematic evidence exists on how ECD teachers experience or respond to thermal stress in their centres.

This constitutes a significant problem for three reasons. First, children in the early years (birth to six) are among the most physiologically vulnerable to heat stress, yet ECD-specific developmental vulnerability to elevated temperatures remains poorly understood and underrepresented in South African environmental health research (Evans, 2006; UNICEF, 2023). Second, teachers in ECD centres are positioned as the first responders to children’s heat-related discomfort, yet their perceptions and coping strategies have not been documented in the South African context. Third, without evidence on how thermal conditions affect teaching and learning in ECD centres, policymakers and infrastructure planners lack the knowledge base to develop appropriate interventions.

This study therefore asks: what do ECD teachers in Hammanskraal perceive to be the impacts of elevated indoor temperatures on children in their centres, and what pedagogical and adaptive responses do they employ? Answering this question is necessary to generate the evidence base required for context-specific ECD thermal comfort policy in South Africa.

1.4 Research Gap and Novelty

The research gap addressed by this study is fourfold:

Geographical: No published study has examined teacher perceptions of thermal stress in ECD centres in Hammanskraal or comparable peri-urban South African communities.

Methodological: Existing thermal comfort research in educational settings relies predominantly on physical measurements and standard comfort models developed for adult occupants (Teli et al., 2012; Wargocki & Wyon, 2017). Qualitative, teacher-centred approaches capturing lived experience in ECD contexts are absent.

Institutional: Studies on indoor environmental quality in South African educational settings have concentrated on primary and secondary schools (Gibberd & Motsatsi, 2013; Mendell & Heath, 2005), not ECD centres — which differ substantially in their infrastructure, occupant vulnerability, and regulatory framework.

Policy: South African ECD infrastructure policy does not currently include thermal comfort standards. This study aims to produce context-specific evidence to inform such standards.

This study therefore claims novelty as one of the first investigations to examine ECD teacher perceptions of thermal stress in peri-urban South Africa, integrating climate change, indoor environmental quality, and early childhood development perspectives to generate evidence capable of informing ECD infrastructure policy and practice.

1.5 Research Purpose

This study will provide a thorough evaluation of the impacts linked to rising temperatures and their possible influence on the children in Early childhood Development (ECD) centres. The research will examine the impacts of higher classroom temperatures on children in ECD centres, with a particular focus on physiological and academic performance indicators. Understanding the unique environmental issues that ECD centres, especially those attended by vulnerable populations, face is crucial as global temperatures rise as a result of climate change. The purpose is to determine the different consequences and provide ways to lessen negative impacts on the children's comfort, wellbeing, and academic achievement. In order to improve the learning environment and develop effective techniques and policies to offset negative impacts.

1.6 Research Questions

This study is guided by the following primary research question:

What are ECD teachers' perceptions of the impact of elevated indoor temperatures on children in ECD centres in Hammanskraal, Gauteng?

The following sub-questions guide the inquiry:

How do ECD teachers perceive the thermal conditions in their centres, and how do these conditions vary across seasons?

What impacts do ECD teachers perceive elevated indoor temperatures to have on children's comfort, behaviour, and participation in learning activities?

What environmental and infrastructural factors do teachers identify as contributing to elevated indoor temperatures in their centres?

What strategies do ECD teachers and centre principals employ to manage thermal stress, and what are the perceived limitations of these strategies?

1.7 Aim and Objectives

a. Aim

The study aims to identify the perceived impact that increased environmental indoor temperature in classrooms has on younger children in Early childhood Development (ECD) centres, in Hammanskraal, Gauteng.

b. Objectives

- To thoroughly review existing literature on the relationship between elevated classroom temperatures and the children's comfort/ well-being.
- To design and implement a robust environmental health assessment methodology tailored to investigate the specific impacts of elevated classroom temperatures on the children, incorporating both qualitative and quantitative research methods.
- To develop evidence-based recommendations and interventions aimed at mitigating the adverse effects of elevated classroom temperatures on the ECD centres' children's wellbeing, encompassing strategies for improved ventilation, thermal insulation, green infrastructure, and climate-responsive building design.
- To evaluate the effectiveness of ventilation and cooling mechanisms in classrooms in mitigating the negative effects of high temperatures on students.

1.8 Justification

This study is justified on scientific, social, and policy grounds. Scientifically, it contributes to a largely absent body of evidence on ECD-specific thermal vulnerability in peri-urban South Africa. The physical environment has been shown to exert profound influence on child development (Evans, 2006), yet thermal quality in ECD centres has received virtually no research attention in the South African context. Studies in the global North have demonstrated that elevated classroom temperatures impair children's cognitive performance and learning (Mendell & Heath, 2005; Wargocki & Wyon, 2017; Teli et al., 2012), but these findings have not been contextualised for ECD settings in semi-arid, resource-constrained, peri-urban African environments.

Socially, the children attending ECD centres in Hammanskraal belong to vulnerable communities with limited access to health services and climate-controlled environments. Understanding the risks they face in their ECD environments is essential for protecting their developmental rights. ECD teachers, as the primary adults responsible for these children during the school day, hold irreplaceable observational knowledge about how heat affects children's wellbeing and learning readiness.

From a policy perspective, South Africa's ECD infrastructure norms do not currently specify thermal comfort requirements. The findings of this study are intended to directly inform the development of such standards by providing empirical, context-specific evidence. This is consistent with the broader imperative to integrate climate change adaptation into early childhood development policy across sub-Saharan Africa (WHO, 2021; Garland et al., 2015).

1.9 Chapter Outline

Chapter One: Introduction. Provides the background, epistemological stance, problem statement, research gap, research questions, aim, objectives, and justification of the study.

Chapter Two: Literature Review. Reviews existing scholarship on climate change in South Africa, thermal stress in educational settings, ECD-specific developmental vulnerability,

teacher perceptions of learning environments, and indoor environmental quality in schools.

Chapter Three: Research Methodology. Details the interpretivist paradigm, mixed-methods design, study area, population, sampling, data collection instruments, data analysis procedures, trustworthiness measures, and ethical considerations.

Chapter Four: Findings and Results. Presents and analyses qualitative findings from teacher interviews and principal interviews, alongside quantitative data from questionnaires and observational checklists.

Chapter Five: Discussion, Conclusions and Recommendations. Interprets findings in relation to existing literature, addresses the research questions, draws conclusions, and offers evidence-based policy and infrastructure recommendations.

CHAPTER TWO: LITERATURE REVIEW

[EDIT: Chapter 2 intro paragraph rewritten to foreground teacher perceptions and ECD-specific framing as requested by supervisor. All original sections retained. New subsections added on ECD infrastructure and teacher perceptions as a lens.]

This chapter provides a detailed review of the literature relevant to the study of teachers' perceptions of elevated indoor temperatures in Early Childhood Development (ECD) centres in Hammanskraal, Gauteng. The review is organised to build from the global and national context of climate change and thermal stress, through to the specific physiological and developmental vulnerability of young children in ECD settings, and the role of teachers as observers and respondents to these conditions. Special attention is given to ECD infrastructure as a compounding factor in thermal exposure, and to the limited but growing body of literature that centres teacher perceptions as a legitimate source of environmental health knowledge. The chapter concludes by identifying the gaps that this study seeks to address.

2.1 Climate Change Around the World

Climate change is a major topic around the world currently and it risks impacting the economy, human health, environment, animal health and a country's development. The history of the climate change discourse reveals that from a purely scientific concern, it has turned into a public agenda that is nowadays more motivated to be a developmental problem, an environmental issue, and according to Chersich et al. (2018), an ecological determinant of health. According to an article written by Lindsey and Dahlman (2023) the Earth's temperature has gone up by an average of 0.08 °C per decade. The rate of increase in temperature has since doubled since year 1981 to 0.18 °C per decade. The year 2022 was recorded as the sixth warmest year ever since the global recordings started in 1880. The global trend observed has indicated that the warming across the globe has not been uniform, although the global average temperature shows that the warming is becoming intense. It is uncertain how much of the warming the Earth will experience, but all is dependent on the emission of greenhouse gases in the coming decades.

The UNICEF (2023) health programme report states that global warming had initiated a lot of extreme events like hurricanes, floods and heatwaves. Exposure to harsh events such as extreme heat and heatwaves is becoming a known experience in most parts of the world. A report by UNICEF confirms that by year 2050 almost every child under the age of 18 in the world, at least 2,2 billion, will be exposed to a rise in heatwave frequency. Also, approximately 559 million children are already under exposure to high levels of heatwaves, and around 624 million children are exposed to one of the three high-heat measures. In the year 2020, approximately 740 million children worldwide were staying in countries with about 83 or more days per year that exceeds 35 °C / 95 F, especially in Africa.

The year 2022 saw most areas across Africa experience extreme events such as heavy rainfall, floods, tropical cyclones, heatwaves, wildfires, droughts and sandstorms. It is recorded that the year 2022 was the fifth warmest year, despite the cooling effect of La Niña. The continent of Africa has recorded warmer days between 1991 and 2022, with an average rate of change of +0,3 °C per decade as compared to the 0,2 °C per decade recorded between 1961 and 1990. Records do confirm that South Africa has seen a rise in mean temperatures of +0,4 °C in 2022, above the average recorded in 1991–2022, World Meteorological Organisation (2023b).

2.2 The South African Climate

The impact of climate change in South Africa on economic growth has been negative and may continue to worsen in the future as predicted. It may severely affect economic growth, creation of jobs, inequality and energy generation. Due to the country's dependency on rain-fed agriculture and natural resources, the impact of climate change in South Africa will be more severe with high vulnerability.

Temperatures have in the past years increased significantly, with temperatures increasing by 1,5 degrees Celsius, with more increases in other parts of the country. The maximum and minimum daily temperatures have increased in all the seasons, with intensity and frequency also increasing significantly across the country World Bank Group (2021). According to a report by the South Africa Weather Services (2021), analysis of long-term

climate data indicates that Gauteng falls within the Cwa region in the north part, while the south part falls under the Cwb region. In Gauteng, extreme heat and temperatures befall during the months of summer. Records indicate that the year 2016 was one of the hottest years globally, with some stations in Gauteng recording the highest temperatures ever, including Unisa Pretoria (42,7 °C), University of Pretoria (39,5 °C) and Vereeniging (38,8 °C).

2.3 Heatwaves in South Africa

The average global temperature has risen as a result of the climate catastrophe, and high-temperature extremes like heatwaves are occurring more frequently. Increased mortality, decreased productivity, and infrastructural damage can all result from higher temperatures. The elderly and young children, who are among the population's most vulnerable, will be the most badly hit European Commission (2024). The South Africa Weather Services (2021) describes heatwaves as a duration whereby there is uncomfortable and abnormal weather that is humid and hot, that may last at least three days or longer.

Heatwaves are consecutive periods of hot days with temperature that is above a given threshold. The rising events of heatwaves around the world can have a severe impact on the health of human beings, from ill health to mortality. The frequency, intensity and duration of the heatwaves are the determining factors related to the severity of the effects of heatwaves. Research has shown that climate change will lead to an increased and frequent extreme weather pattern. The projections for climate change in Africa has indicated that there is an increase in frequency, duration and intensity of heatwaves, which all have a potential to negatively impact human health, air quality and environmental conditions, Kapwata *et al* (2022).

Statistics and records for South African climates indicate that there has been a significant annual trend of warm days increasing at the rate of 0,15% annually and the occurrence of heatwaves increasing by 0,03% event per year in the same period. In South Africa (SA), it has been recorded that the highest frequency of heatwaves occurred during the austral summer with 150 out of 270 events between the years 2014 and 2019. The

intensity and duration of the heatwaves that were recorded inland in SA have a potential to increase the risk of heat-related mortality, Kapwata et al (2022). Hammanskraal, like most areas in the Tshwane Municipality, has experienced unusual weather patterns in both the winter and summer seasons, making it a particularly relevant context for examining how thermal stress affects the youngest and most vulnerable occupants of educational settings.

2.4 Physiological Impacts of Increased Heat on Younger Children

No Changes

2.5 ECD Centres: Infrastructure and Thermal Vulnerability

[EDIT: New section added as requested by [REDACTED] and to address supervisor's comment that environmental design discussion dominates but lacks ECD-specific institutional focus.]

Early Childhood Development centres in peri-urban South Africa, including those in Hammanskraal, often operate in conditions that differ substantially from formal primary or secondary schools. Many ECD centres are housed in converted residential structures, community halls, or purpose-built but under-resourced facilities that were not designed with thermal comfort or child safety in mind (Garland et al., 2015). The infrastructure challenges in these centres are well documented in the South African context, though thermal performance has received limited dedicated attention.

The materials commonly used in the construction of ECD centres in Hammanskraal are a significant contributor to elevated indoor temperatures. Corrugated iron roofing, which is the most prevalent roofing material in informal and peri-urban educational facilities, absorbs solar radiation rapidly and transfers heat into the interior of the building, raising indoor temperatures to levels that can exceed outdoor temperatures by 2 to 4 °C (National Department of Health, 2020). The absence of ceiling insulation further eliminates the buffer between the iron roof and the indoor learning space, resulting in direct radiant heat transfer onto the children and teachers below. Gibberd and Motsatsi (2013), in their study of a classroom in Atteridgeville, Pretoria, found that indoor air temperatures closely

mirrored outdoor air temperatures and that the building envelope offered negligible thermal comfort protection.

In addition to roofing, plastered brick walls; the most common exterior wall material in the centres studied, absorb heat during the day and release it slowly at night, contributing to elevated indoor temperatures during and after school hours. The absence of window shading, including blinds, curtains, or external overhangs, allows direct solar radiation to enter classrooms and further raise the radiant temperature experienced by children. The absence of mechanical cooling in the overwhelming majority of ECD centres in Hammanskraal means that natural ventilation (primarily through open windows and doors) is the only available temperature management strategy available to teachers.

South Africa's ECD infrastructure policy, as governed by the Norms and Standards for ECD Programmes under the Children's Act, does not currently prescribe thermal comfort standards or minimum ventilation requirements for ECD centres. This policy gap means that even centres that are formally registered and inspected may be operating in thermally hazardous conditions without any regulatory requirement to address them. This absence of ECD-specific thermal guidelines represents a significant institutional gap that this study seeks to contribute to addressing, by generating teacher-centred evidence that can inform policy development.

2.6 Teacher Perceptions as a Lens for Understanding Thermal Stress

[EDIT: New section added to address supervisor's core comment that teacher perceptions are underdeveloped throughout the thesis. This section establishes the theoretical justification for centring teachers' perceptions.]

The study of teacher perceptions as a source of environmental health knowledge is grounded in an interpretivist understanding of how expertise is constructed in educational settings. Teachers in ECD centres are not merely transmitters of curriculum; they are sustained observers of children's behaviour, health, and wellbeing across the full school day. Their ability to notice and interpret changes in children's attention, mood, physical comfort, and participation constitutes a form of observational expertise that is both contextually grounded and practically significant.

Bryman (2016) argues that in interpretivist research, the subjective meanings that participants assign to their social world are legitimate sources of knowledge, particularly when the phenomena under investigation are experienced rather than measured. In the context of thermal stress in ECD centres, teachers' perceptions of how heat affects children are not inferior substitutes for instrumented temperature data; they are a different and complementary form of evidence that captures the lived consequences of environmental conditions. A teacher who observes that children become irritable, lose concentration, sweat excessively, or ask for water more frequently during hot days is providing data that no thermometer can capture.

This epistemological position is consistent with the mixed-methods design of this study, which combines qualitative data from teacher interviews and principal interviews — capturing perceptions, experiences, and interpretations — with quantitative data from questionnaires and observational checklists. The integration of these data sources allows the study to document both the thermal conditions in ECD centres and the human experience of those conditions, producing evidence that is relevant to both infrastructure planning and teacher support policy.

Studies in the broader field of indoor environmental quality have increasingly recognised the value of occupant perceptions alongside physical measurements. Teli, Jentsch and James (2012) demonstrated in a field study of UK primary schools that children's thermal comfort perceptions diverged significantly from predictions made by standard adult-based comfort models, arguing for perception-based data as an essential complement to instrumented assessment. Mendell and Heath (2005) similarly concluded, in their comprehensive review of indoor environmental quality and student performance, that teacher and student-reported experiences are among the most reliable indicators of classroom conditions that impair learning. These findings support the decision of this study to position teacher perceptions as a primary source of evidence about thermal stress in ECD centres.

2.7 Academic and Cognitive Impacts of Increased Heat on Children

In addition to physical risks, heat can hinder learning. There is a correlation between decreased cognitive function and academic accomplishment and exposure to higher temperatures. In schools without air conditioning, every degree Fahrenheit increase in the average school year temperature decreased the amount of knowledge children learned by 1%, according to research involving 10 million PSAT takers Ahdoot et al (2024).

Heat is linked to decreased cognitive function, slower reaction times, and less capacity to focus. Numerous studies have demonstrated that pupils perform worse in classrooms with higher temperatures. During a six-day intense heat wave in the United Kingdom, a study found that children slept fewer hours, woke up more frequently, took longer to fall asleep, and received more visits from parents at night. High temperatures might interfere with sleep cycles, making children tired and irritable. Extreme heat or cold might interfere with a child's ability to sleep comfortably, which can negatively impact the child's general health and cognitive development Boo et al (2024). Wargocki and Wyon (2017) confirmed that reducing classroom temperatures from 30 °C to 20 °C is projected to improve student performance by approximately 20% on average — a finding that has direct implications for ECD centres operating without any temperature control during summer months.

2.8 Emotional and Behavioural Impacts of Increased Temperatures

Schoolchildren who are exposed to severe temperatures for extended periods of time may experience stress and anxiety. Children can become more stressed and find it harder to concentrate on their studies if they worry about being too hot, uncomfortable, or unable to concentrate in class. Prolonged stress can have a detrimental effect on mental health and academic performance. The stress response mechanism in children's brains is triggered when temperatures are high. Overstimulation of the stress response system during this vulnerable period can interfere with the development of emotional regulation circuits, which are essential for the person's health and social interaction Boo et al (2024).

Uncomfortably high temperatures might make the classroom difficult. When children are too hot, they get fidgety, agitated, and less interested in the lessons being taught. When students are uncomfortable, it can also be difficult for teachers to keep the classroom

under control and organised, which can cause disruptions and lower productivity Lala & Hagishima (2023).

From the perspective of ECD teachers, these behavioural changes are particularly challenging to manage because young children in the birth-to-six age range lack the cognitive and verbal capacity to clearly communicate their discomfort. Teachers in ECD centres must therefore read behavioural cues — increased crying, irritability, reduced participation, excessive physical restlessness, requests for water, or unusual drowsiness — as indicators of thermal stress. This makes ECD teachers' observational perceptions an especially important data source in settings where children themselves cannot self-report.

2.9 School Buildings and Their Infrastructure

According to Smith (2023), with the increase in heat waves across the world, there are heightened urban temperatures that may result in the development of increased risk for heat-related illnesses in urban residents due to urban heat islands. These may be enhanced by concentrations of pavements, buildings, and other surfaces that absorb and retain heat, elevating the ambient temperatures in cities by up to 12.2 degrees Celsius.

There is a need to review the types of buildings used as educational facilities around the world and the materials used to build them. This would assist in identifying if the buildings are contributing to the children's exposure to extreme heat while at school. Tagliabue et al (2020) states that educational buildings in European countries mostly consist of a large number of structures that currently require crucial interventions. The old buildings require an essential makeover to improve the indoor comfort conditions, creating a suitable environment for students. The study confirms that during the season of summer, the indoor thermal comfort in school buildings is a serious issue in different areas of Europe.

The relevance of global findings on educational building performance to the ECD context in Hammanskraal lies in the shared principle that building materials and design choices directly mediate children's thermal exposure. However, the South African ECD context differs from the European school building literature in two important respects. First, the resource constraints in peri-urban ECD centres mean that the intervention options

identified in higher-income contexts; such as retrofit insulation, mechanical cooling, or green building systems, are often financially inaccessible. Second, unlike formal schools, ECD centres in South Africa are not subject to the same infrastructure norms and standards, meaning that baseline quality requirements are lower and less consistently enforced. These distinctions underscore the importance of generating South Africa-specific, ECD-specific evidence about the relationship between building infrastructure and thermal comfort — precisely the kind of evidence that this study, through teacher perceptions and observational data, seeks to provide.

2.10 Chapter Summary

This chapter has reviewed the literature relevant to understanding thermal stress in ECD centres through the lens of teacher perceptions. The evidence confirms that climate change is intensifying heat stress across South Africa, that Hammanskraal is situated in a climatic zone particularly susceptible to extreme temperatures, and that young children bear the greatest physiological burden of heat exposure due to their underdeveloped thermoregulatory systems and dependence on adult caregivers. The literature further reveals significant gaps: existing studies have focused predominantly on formal schools rather than ECD centres, on physical measurements rather than teacher perceptions, and on global or national contexts rather than the peri-urban South African reality. This study is positioned to address each of these gaps. Chapter Three details the methodology through which teacher perceptions were collected, analysed, and interpreted to produce evidence relevant to ECD thermal comfort policy in South Africa.

CHAPTER THREE: METHODOLOGY

[EDIT: Chapter 3 original content retained in full. Epistemological positioning paragraph added at the start as required by supervisor. Cross-reference with Chapter 1 added. Slight framing edits to align study design section with declared mixed-methods/pragmatist paradigm.]

3.1 Introduction

This chapter details the methodology that was used to test the study's objectives empirically and to explore the perceptions of ECD teachers and principals regarding the impact of elevated indoor temperatures on children in Hammanskraal. This section includes a discussion of the epistemological positioning of the study, the research instruments and sampling techniques used, the research population that participated, and the data collection method that was used. The issues of data reliability, trustworthiness, and ethical considerations are also elaborated upon.

3.2 Epistemological and Ontological Positioning

[EDIT: New section added to address supervisor's comment that the epistemological stance is unclear at PhD level. Aligned with Chapter 1 Section 1.3.]

As stated in Chapter One, this study is grounded in a pragmatist epistemological position, which accommodates the mixed-methods design employed. Pragmatism holds that the most appropriate research approach is the one best suited to answering the research questions at hand, and that qualitative and quantitative methods can be integrated when both are necessary to generate a complete understanding of the phenomenon under investigation (Creswell & Poth, 2018). This position is appropriate for the present study because the research questions require both the documentation of physical environmental conditions in ECD centres (addressed through quantitative questionnaires and observational checklists) and an understanding of how teachers perceive, interpret, and respond to those conditions (addressed through qualitative interviews and open-ended questions).

Ontologically, the study acknowledges that thermal conditions in ECD centres are real and measurable, but that the impact of those conditions on children is mediated by context, perception, and the adaptive responses of teachers. The knowledge produced by this study is therefore not limited to what thermometers record; it encompasses what teachers observe, how they interpret children's responses to heat, and what actions they take in response. This positioned the study at the interface between physical exposure and lived pedagogical experience; an interface that requires both quantitative and qualitative methods to investigate adequately. The qualitative component of the study was approached from an interpretivist stance, in which teachers' perceptions are treated as valid and meaningful knowledge that reflects their sustained, contextually embedded observations of children in thermally stressed environments (Bryman, 2016).

This epistemological positioning has practical implications for data collection and analysis. It means that when teachers report that classrooms are uncomfortably hot, that children become irritable or lose concentration, or that outdoor activities are curtailed due to heat, these reports are treated as substantive findings rather than subjective impressions. The thematic analysis of interview data and the statistical analysis of questionnaire responses are both understood as ways of constructing knowledge about a shared reality; the thermal conditions in ECD centres in Hammanskraal and their impact on the children and teachers who inhabit those spaces daily.

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CHAPTER FOUR: RESULTS AND DISCUSSION

[EDIT: Chapter 4 original content retained in full. Introduction and discussion section framing slightly edited to consistently foreground teacher perceptions as the study's primary lens, as required by supervisor. Connecting sentences added to strengthen alignment between findings and the study's focus.]

4.1 Introduction

This section presents the results from the analysis of data collected in various Early Childhood Development centres in Hammanskraal, Pretoria, on the impact of elevated indoor temperatures on children in ECD centres. The chapter is structured to present findings in relation to the study's core focus: the perceptions and lived experiences of ECD teachers and principals regarding thermal stress in their centres.

The findings are organised to address the study's research questions progressively: beginning with the demographic profile of the study sample, moving through teachers' perceptions of classroom climate conditions, the observed impacts of heat on children's health and behaviour, the effects on learning performance, and the mitigation strategies employed by teachers and centre management. The chapter concludes with observational findings on the physical infrastructure of ECD centres, which contextualise and support the patterns identified in the teacher and principal data. Throughout the discussion, findings are interpreted in relation to the literature reviewed in Chapter Two, with particular attention to the ways in which teachers' perceptions both confirm and extend the existing evidence base on thermal stress in educational settings.

4.2 Response Rate

The survey targeted a total sample size of 57 teachers and 12 principals, which was an adequate sample to give a 95% confidence level with 5% marginal error using the Yamane formula. The returned responses were exactly equal to the targeted number, with 1,8% male participants and 98,2% female participants. The 100% response rate is

indicative of the high level of engagement from teachers and principals in the study, and may reflect the relevance of the research topic to their daily professional reality.

CONTINUE TO 4.3 INSTRUMENT RELIABILITY TEST

4.5 Climate and Classroom Conditions

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This section details the classroom conditions in the ECDs as perceived by teachers and the climate conditions they have observed in the area. It is important to note that the data presented here reflects teacher perceptions; their observational and experiential knowledge of the thermal environment in their centres; rather than instrumented measurements. This is consistent with the study's design, in which teacher perceptions are positioned as the primary lens through which the impact of elevated temperatures on children is understood.

4.5.1 Describing the Observed Typical Climate Condition by Teachers: The study requested that participants indicate what the observed typical climate conditions were inside the classrooms during hot days. At least 37 participants (65%) were able to confirm that the classrooms are very hot during summer season, while 19 participants (33%) identified the classrooms as having dry conditions as the temperature increases. Also, 14 participants (25%) responded that it is merely hot, while 6 participants (11%) thought it is warm. The survey concludes that in most hot days, the ECDs experience extreme heat conditions, which may have an impact on how the children experience the indoor environment. A study by Gibberd & Motsatsi (2013) confirms that factors such as ventilation, climate change and infrastructure are likely to increase the chance of exposure to high temperatures in the classrooms.

4.5.2 Challenges Related to Heat Observed in Classrooms: A total of 41 out of 57 (72%) participants said they have identified challenges related to heat in the classrooms, while 15 (26%) said no and only 1 (2%) said they were not sure. The national guidelines by National Department of Health (2020) confirm that indoor temperatures in structures

made of steel, painted dark colours, or with little or no heat insulation in the roof or walls can be about 2 to 4 °C warmer than outdoors. The guidelines explain how some building materials and designs can cause interior temperatures to be higher than exterior temperatures, resulting in various heat-related challenges in ECDs.

4.5.3 Different Challenges as Observed by the School Teachers: When asked to specify the challenges they have encountered or observed in the classroom, 30% of participants confirmed that the classrooms are becoming warmer, 16% confirmed restricted or cancelled outdoor activities due to excessively high temperatures, 20% said learners become sick or sleepy during hot days, and 7% said the children tend to sweat excessively. Majority of the participants (35%) preferred not to disclose the challenges that were observed in the classroom. The results suggest that the impact of heat is affecting both the comfort of children and the learning environment in the classrooms.

4.6 The Impact of Increased Heat on the Children

Majority of participants (35%) believed that the rooms became uncomfortable and too hot for the children. A portion of 21% thought that the children lacked concentration during lessons when it is too hot, 11% said the children are always tired, and 7% said they sweat a lot. The teachers have noted a significant discomfort on the children when it is hot in the classrooms. These findings align with a study done by Park et al (2020) and Goodman et al (2019) where they proved that high temperatures in classrooms are linked to low academic performance of children leading to lower test results.

The perceptions of ECD teachers in this study offer a particularly important perspective on these impacts because they are drawn from sustained, day-to-day observation of very young children who cannot self-report. The fact that 90% of participants observed a change in behaviour or attention during hot days — with irritability and loss of concentration being the most commonly reported changes — is consistent with the physiological evidence reviewed in Chapter Two. Children's brains divert cognitive resources to thermoregulation during heat stress, reducing the capacity available for learning, attention, and emotional regulation (Early Childhood Council on Equity & Environment, 2023). That ECD teachers are observing these changes consistently across

different centres and with different children confirms that the phenomenon is real, widespread, and significant.

4.7 Health Concerns

A total of 88% of participants confirmed that the learners usually need more water to drink on hot days, while 53% observed that children looked restless and 51% confirmed an observation of excessive sweating. Based on the findings, this confirms that the heat exposure could have negative health impacts on younger children. A study by Bidassey-Manilal et al (2016) conducted in various high schools in South Africa found that most students experienced fatigue and low concentration during hot days, with temperature ranges above 32 degrees Celsius directly linked to the reported feeling of tiredness and struggling to breathe.

4.8 Learning Performance

Participants were asked if they think the increased heat in the classroom has any impact on the performance of the learners. A total of 60% said yes, while 40% said they do not think there is an impact. They were further asked if they have noted any impact of heat on learners' performance during lessons. A total of 63% said yes, they have observed a change in performance while 37% said they have not noted any change. Studies conducted by Goodman et al. (2019) and Park et al. (2020) confirm that there is a global trend where reduced cognitive performance and academic achievement is linked to high exposure to elevated temperatures.

The learning performance findings in this study are particularly significant when considered in the context of ECD. Early childhood is recognised as the most critical period for cognitive and socio-emotional development, and disruptions to learning readiness during this period can have long-lasting developmental consequences (Evans, 2006). The fact that 63% of ECD teachers observed changes in learning performance during hot days — including reduced concentration (39%), sleepiness (9%), restlessness (5%), and withdrawal from activities (4%) — suggests that thermal stress in ECD centres is not merely a comfort issue but a developmental one. These findings have direct implications

for early childhood education policy, infrastructure planning, and teacher training in South Africa.

4.9 Strategies and Mitigation

Participants were asked about the strategies and mitigation measures that they had in place in their respective ECDs to deal with the impact of heat on the learners. A total of 88% agreed that opening windows and doors helped by reducing indoor heat. A total of 77% also agreed that giving the learners water frequently helped with keeping them hydrated. Then 72% of the respondents believed that making the children sit outside under shades helped with easing the impact of heat. Only 28% of the participants noted that they used mechanical ventilation such as fans and air conditioners.

The predominance of low-cost, behavioural mitigation strategies — opening windows, providing water, moving children outdoors, adjusting schedules — reflects the resource constraints of ECD centres in Hammanskraal and the limited infrastructure available to teachers. These findings also reveal the extent to which ECD teachers have developed practical, adaptive responses to thermal stress based on their direct observational experience of what works for the children in their care. That teachers are the primary architects of heat management in these centres — in the absence of institutional infrastructure, policy guidance, or financial support — underscores the central importance of teacher perceptions in understanding how thermal stress is managed in ECD settings. At the same time, the limitations of these strategies are clear: natural ventilation and hydration can moderate but not eliminate the thermal stress imposed by corrugated iron roofing, absent ceilings, and no mechanical cooling, particularly during peak summer temperatures in Hammanskraal.

4.10 Emotional Observation by Teachers

Participants were asked to rate the observed emotions of the learners during hot days. Majority of the respondents (26%) strongly agreed that the learners appeared generally comfortable despite the high temperatures, while 40% strongly agreed that learners seemed restless or irritable during lessons on hot days, and 26% agreed. The results

validate that high indoor temperatures in Hammanskraal ECD centres present serious health and educational issues.

Teachers have suggested both short-term (hydration, shade, flexible schedules) and long-term (fans, better building design, larger classrooms) solutions in response to these issues. Therefore, addressing classroom heat is essential to defending children's rights to a good education and general wellbeing. According to Pouloupoulos (2025), higher temperatures have been related in studies to more mood fluctuations, disturbed sleep, and decreased focus. Heat can exacerbate pre-existing issues for children who already have anxiety, emotional management challenges, or ADHD, which can result in more tantrums, arguments, and reluctance to routines.

4.11 Observation of the Infrastructure in the ECDs

To evaluate the physical features of the ECD classrooms' infrastructure in Hammanskraal, the observation checklist was employed. The findings illuminate the ways that design elements and construction materials can raise indoor temperatures and cause general thermal discomfort in the classroom.

Roof Material: A total of 88% of the centres used corrugated iron roofing, 10% used clay tiles while 2% used asbestos roofing. The high number of corrugated iron roofing indicated that most classrooms are gaining a lot of heat during hot seasons. A lot of studies around the world have shown that corrugated iron roofing and other metal roofing materials have been repeatedly linked to elevated thermal stress in classrooms (Gibberd & Motsatsi, 2013; Nyamadzawo et al., 2020).

Ceiling Material: A total of 58% did not have any ceiling installed in their centres, 40% used ceiling boards while 2% used wooden ceiling. The results indicate that majority of the centres did not have any ceiling installed in their classrooms, exposing the learners to direct radiant heat from the iron roofing. Research has demonstrated that ceilings are essential for controlling indoor temperatures by preventing direct heat transfer (UNICEF, 2021; South African Green Building Council, 2020).

Temperature Controls: A total of 84% of the centres did not have any temperature control systems and only 16% had. The few centres that had a temperature control system used fans and air conditioners to cool down the classrooms during hot days. This result is consistent with teachers' reports that during hot days, temperatures in classrooms are very high.

The infrastructural findings of this study provide the physical context that explains and validates the teacher perception data presented in earlier sections of this chapter. Teachers who report that classrooms are 'very hot' during summer are not exaggerating; they are describing the predictable consequence of corrugated iron roofing, absent ceiling insulation, unshaded windows, and no mechanical cooling. The convergence of the infrastructural observations (collected by the researcher) and the teacher perception data (collected through questionnaires and interviews) constitutes a form of methodological triangulation that strengthens the credibility of the study's findings. It confirms that the thermal stress perceived by teachers, and the behavioural and health impacts they observe in children, are grounded in real and documentable physical conditions.

These findings have direct implications for ECD infrastructure policy in South Africa. The evidence from this study supports the urgent need for ECD-specific thermal comfort standards and infrastructure guidelines that reflect the particular vulnerabilities of the under-six age group, the resource constraints of peri-urban ECD operators, and the South African climate context. Such guidelines should, at a minimum, address roofing materials, ceiling insulation requirements, window shading, natural ventilation design, and access to basic cooling resources. The knowledge and experience of ECD teachers — as documented in this study — should be treated as a core input into the development of these guidelines.

4.12 Chapter Summary

This chapter has presented and discussed the findings of the study across all data collection methods. The teacher perception data, principal interview data, and infrastructural observational data converge consistently around a central finding: elevated

indoor temperatures in ECD centres in Hammanskraal are a real, pervasive, and consequential phenomenon that affects the health, behaviour, comfort, and learning performance of young children. ECD teachers, as the primary observers and managers of this phenomenon, possess substantive, contextually grounded knowledge about its impacts and about the strategies that help children cope. This knowledge is currently underutilised in South African ECD policy and infrastructure planning. Chapter Five presents the conclusions drawn from these findings and the evidence-based recommendations that the study offers to address this gap.